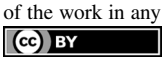
Micro-Electrochemical Properties of CeS Inclusions in Stainless Steel and Inhibiting Effects of Ce^{3+} Ions on Pitting

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Micro-scale polarization data from small surface areas of stainless steels with either CeS or MnS inclusions was measured in NaCl and Na_2SO_4 solutions to elucidate the dissolution and pit initiation behavior of both types of inclusions. Stable pitting was initiated at the CeS inclusion at potentials which exceeded the dissolution potential region of the CeS inclusion, whereas a stable pit at the MnS inclusion was initiated in the dissolution potential range of the MnS inclusion. Thermodynamic calculations indicated that the Ce^{3+} ions are likely to be produced by the dissolution of the CeS inclusion. In the micro-scale polarization of a small area with a MnS inclusion, stable pitting at the MnS inclusion was inhibited in NaCl solutions containing Ce^{3+}. The formation of a shallow trench was observed at the MnS inclusion/steel matrix boundary in the Ce^{3+}-containing solution, whereas a deep trench was formed in the Ce^{3+}-free solution. It is suggested that the Ce^{3+} ions inhibit trench formation at the MnS/steel matrix boundary, resulting in improved pitting corrosion resistance at sulfide inclusions.

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While stainless steels are successfully used in corrosive environments, they sometimes suffer from pitting. Improving the pitting corrosion resistance of stainless steels is of the utmost importance because of the vital role these steels play in infrastructure, transportation, energy, in bridges, and buildings in general. The pit initiation sites for commercial stainless steels have been attributed to sulfide inclusions such as MnS in Cl^--containing environments,[1–10] and hence considerable research has been carried out to clarify the mechanism of the pitting at MnS inclusions. It has been proposed that the initial step of the pit initiation process at MnS inclusions is the anodic dissolution of the inclusions and that the complete dissolution of the inclusions is not required for pit initiation.[11–21] The release of S-containing species during MnS dissolution increases pitting susceptibility at and around the inclusions. Webb and Alkire demonstrated that the dissolution of the inclusions produces $S_2O_3^{2-}$ ions and leads to weak acidification near the inclusions.[14] Chiba et al. proposed that the synergistic effect of chloride ions and dissolution products from MnS causes trench formation at the boundary of the inclusion/steel matrix, and pitting occurs at MnS inclusions as a local transition from the passive to the active state of the steel matrix inside the trench.[19] The effect of minor elements on the pitting at MnS inclusions has been investigated. Lillard et al. proposed that Cu ions are released from the steel matrix inside the trench and MnS inclusions are ennobled by Cu deposition.[21] They suggested that the ennobled MnS inclusions accelerate pit propagation inside the trench. It is therefore reasonable to assume that the inhibition of the dissolution of the inclusion will directly suppress the formation of trenches, resulting in an improvement in the pitting corrosion resistance of stainless steels.

The chemical composition of sulfide inclusions has a significant influence on pitting corrosion behavior. It has been reported that Cr-sulfide, Ti-sulfide, and Ti-carbosulfide inclusions have better pitting corrosion resistance than MnS inclusions.[22–30] Muto et al. and Shimahashi et al. revealed that the oxide-enriched layers generated on the inclusion surfaces prevent the anodic dissolution of the inclusions and improve pitting corrosion resistance.[25,29] It is also reported that Ce-sulfide and oxysulfide inclusions have higher pitting corrosion resistance than MnS inclusions.[31–34] Because it is a rare-earth element, Ce is relatively reactive in aqueous environments. While the pitting corrosion resistance of CrS and TiS inclusions is attributed to the thermodynamic stability of the surface oxide films on the inclusions, the stability of the surface oxides on CeS inclusions is unlikely to be higher than that of CrS or TiS. The pitting corrosion resistance of CeS inclusions is therefore the result of other factors and is possibly related to the Ce-species produced by the dissolution of the inclusions.

Cerium has been shown to be effective in the corrosion inhibition of Al alloys.[35–40] Hinton et al. indicated that the corrosion rate of Al alloys decreased in NaCl solutions with the addition of Ce salts.[35] They found that the oxygen reduction reaction is inhibited by the formation of Ce-containing oxide layers on the cathodic sites. The corrosion inhibition for stainless steels by Ce-species has also been investigated.[41–45] Lu and Ives demonstrated that Ce chemical treatment improves the crevice corrosion resistance of stainless steels.[41,42] Also in the case of stainless steels, Ce chemical treatment is expected to inhibit the cathodic reactions. It is therefore predicted that Ce-species released from CeS inclusions affect the pitting corrosion resistance of stainless steels. However, little is known about the electrochemical properties of CeS inclusions, and the relationship between those properties and pitting corrosion resistance remains unclear.

In this paper, the micro-scale polarization of small surface areas of stainless steels with either CeS or MnS inclusions was measured in NaCl and Na_2SO_4 solutions to elucidate the dissolution and pit initiation behavior of the inclusions. The potential-pH diagram for the Ce-S-CeS-H_2O system was calculated to determine the dissolution products from the CeS inclusion. The effect of Ce ions on the pitting corrosion resistance at MnS inclusions was investigated in NaCl-$CeCl_3$ solutions. Finally, the role of Ce ions in improving the pitting corrosion resistance of the sulfide inclusions is discussed based on the micro-scale polarization and the cross-sectional morphology of the inclusions after polarization.

Experimental

Specimens.—Two types of stainless steels were prepared by vacuum induction melting (50-kg ingot) and then were hot-rolled. The chemical compositions of the steels are listed in Table I. In steel A, Mn and S were added to form MnS inclusions. In steel B, Ce and S were added to form CeS inclusions. These steels were not deoxidized by Al to avoid the formation of Al-rich inclusions.[46] The steels were cut into specimens with dimensions of 15 mm × 25 mm × 5 mm. All the specimens were heat-treated at 1373 K for 0.5 h and quenched in water. After heat-treatment, the specimens were abraded with a series of SiC papers up to 1500 grit and then polished by a diamond paste down to 1 μm. The specimens were finally rinsed with ethanol. Since the non-metallic inclusions are darker in color under an optical microscope compared to the stainless steel matrix, they are clearly visible in the as-polished condition without etching.

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Table I. Chemical compositions of specimens (mass%).

Steel	C	Si	Mn	P	S	Ni	Cr
A	0.051	0.39	1.51	0.039	0.0215	8.27	18.3
B	0.011	0.47	<0.05	0.033	0.0170	9.41	17.9
Steel	Mo	Cu	Al	Ce	N	O	
A	0.21	0.35	0.002	–	0.0799	0.0023	
B	<0.01	<0.01	<0.005	0.091	0.0039	0.0080	

–: No addition.

Micro-scale anodic polarization.—Potentiodynamic anodic polarization was carried out in 0.1 M NaCl and 0.1 M Na_2SO_4 solutions. The pH values of the 0.1 M NaCl and 0.1 M Na_2SO_4 solutions were 5.5 and 5.8, respectively. In addition to these solutions, NaCl-$CeCl_3$ and Na_2SO_4-$Ce_2(SO_4)_3$ solutions were also used. The chemical compositions and pH values of the solutions are shown in Table II. The solutions were prepared from deionized water and analytical grade chemicals. All measurements were performed at 298 K under naturally aerated conditions. To make a micro-scale electrode area containing a single inclusion, the specimen surfaces were masked by a vinyl copolymerized resin (Sunhayato VA-30). The electrode area was ca. 100 μm × 100 μm. Since this resin is hydrophilic, it is easy that solution was placed on the specimen without leaving air bubbles on the electrode surface. An acrylic cell was placed on the masked specimen to preserve the solution. A small Ag/AgCl reference electrode and a Pt wire counter electrode were fixed to the acrylic cell. All the potentials reported in this study refer to the Ag/AgCl (3.33 M KCl) electrode (0.206 V vs. standard hydrogen electrode at 298 K). A battery-powered potentiostat[17] was used to control the electrode potential of the specimen with a scanning rate of 3.83×10^{-4} V s^{-1} (23 mV min^{-1}). Although slower scan rates are reliable for potentiodynamic polarization in general (e.g., 1.67×10^{-4} V s^{-1} (10 mV min^{-1})),[47] the higher scan rate was used to shorten the experiment time and to reduce the data capacity of the in situ observation system described below.

During polarization, the electrode area was observed by the in situ real-time optical micro-scale observation system. The details of the system developed by Chiba et al. have been published elsewhere.[18] The micro-electrochemical cell was put on the stage of an Olympus BX51M optical microscope. An Olympus LUMPlan Fl 100XW water immersion objective lens was used to obtain the real-time video images of the micro-scale electrode surface during polarization.

Observation of surface and cross-section of specimens.—An optical microscope (OM) and a Keyence VK-8510 confocal laser scanning microscope (CLSM) were used to image the surface of the electrode area and the inclusions before and after micro-scale anodic polarization. A field emission scanning electron microscope (FE-SEM) equipped with an energy-dispersive X-ray spectroscopy (EDS) system was used to observe the surfaces and determine the chemical compositions of the inclusions. The accelerating voltage was 20 kV. Cross-sections of the MnS inclusions after micro-scale anodic polarization were prepared by a focused ion beam (FIB) system. The secondary electron images of the cross-sections were taken at an electron accelerating voltage of 10 kV.

Table II. Chemical compositions and pH values of solutions.

Solution	NaCl [M]	$CeCl_3$ [mM]	Na_2SO_4 [M]	$Ce_2(SO_4)_3$ [mM]	pH
1A	0.1	–	–	–	5.5
1B	0.097	1	–	–	5.4
1C	0.07	10	–	–	5.3
2A	3	–	–	–	5.0
2B	2.997	1	–	–	5.0
2C	2.97	10	–	–	5.0
3A	–	–	0.1	–	5.8
3B	–	–	0.0985	0.5	5.8
3C	–	–	0.085	5	6.0

–: No addition.

Titrations.—To confirm whether the Ce^{3+} ions are a pH buffer or not, the titration curves of the NaCl-$CeCl_3$ solutions were measured. The test solutions were 0.1 M NaCl (pH 5.5), 0.097 M NaCl-1 mM $CeCl_3$ (pH 5.4), and 0.07 M NaCl-10 mM $CeCl_3$ (pH 5.3). These test solutions were prepared from deionized water and analytical grade chemicals. The NaCl-$CeCl_3$ solutions were titrated by adding 0.1 M HCl, and the pH of the solutions was measured.

Results and Discussion

Inclusion characterization.—The chemical composition and morphology of the inclusions were characterized using an optical microscope and the FE-SEM/EDS. Figure 1 shows the optical micrographs, SEM images, and EDS maps of the inclusions. Table III shows the results of the EDS analysis at each point in Fig. 1. In steel A (the Ce-free specimen), as shown in Fig. 1a, the inclusion was elongated and the surface was gray in the optical micrograph. On the other hand, in Fig. 1b, the inclusion in steel B (the Ce-containing specimen) was round and the surface was orange and dark gray, with light gray areas in the optical micrograph. The chemical composition of the inclusions affects their morphology and surface color. In Fig. 1a, enriched Mn and S were present in the inclusion in steel A. The relative atomic percentage of Mn:S:Cr:O at point "A" was 48:47:5:<1, indicating the formation of the inclusions composed of Mn, S, and a small amount of Cr (approximately 5 at.%). This is the typical sulfide inclusion in commercial type 304 stainless steels and is referred to as MnS in this paper.

In the optical micrograph of steel B, the inclusion consisted of orange, dark gray, and light gray areas. The orange and dark gray areas corresponded to the bright area (points "B1" and "B2") in the SEM image. The light gray area (point "B3") in the optical micrograph was darker than points "B1" and "B2" in the SEM image. The EDS maps show that the orange and dark gray areas ("B1" and "B2") were rich in O, and the S content was relatively low. In contrast, the enrichment of S and a small amount of Cr and O was detected in the light gray area ("B3"). As Table III shows, the relative atomic percentage of Ce:S:O at point "B1" was 41:16:43. There was little difference in the chemical compositions between point "B1" and "B2". It was suggested that the orange and dark gray areas in the optical micrograph were Ce_2O_2S (cerium oxysulfide). It is possible that small difference in the O content between point "B1" and "B2" led to the different colors of the Ce_2O_2S inclusion. It has been reported that the color of cerium oxide varies with the oxidation state.[48] The light gray area in the optical micrograph was mainly composed of Ce and S even though small amounts of Cr and O were contained in the inclusion. The relative atomic percentage of Ce:S:Cr:O at point "B3" was 32:48:15:5, indicating the formation of complex sulfide of Ce and Cr. There was a small enrichment of Ca in the right edge of the light gray area. For simplicity, the light gray area of the inclusion in steel B is referred as CeS in this paper. The Ce_2O_2S is thought to be a deoxidation product, and the CeS was produced on the Ce_2O_2S surface in the solidification process of the steel. Since Al was not added to avoid the formation of Al-rich inclusions, Ce acted as a deoxidation agent. CeS could be distinguished from Ce_2O_2S without EDS analysis because the different colors of CeS and Ce_2O_2S made it possible to visually discern one from the other in the optical micrographs.

Polarization behavior of MnS inclusion.—Micro-scale anodic polarization was conducted to confirm the pit initiation behavior at the MnS inclusion. Figure 2a shows the polarization curve of a small area with the MnS inclusion in steel A measured in 0.1 M NaCl. Anodic polarization was started at −0.2 V and was stopped at 0.41 V.

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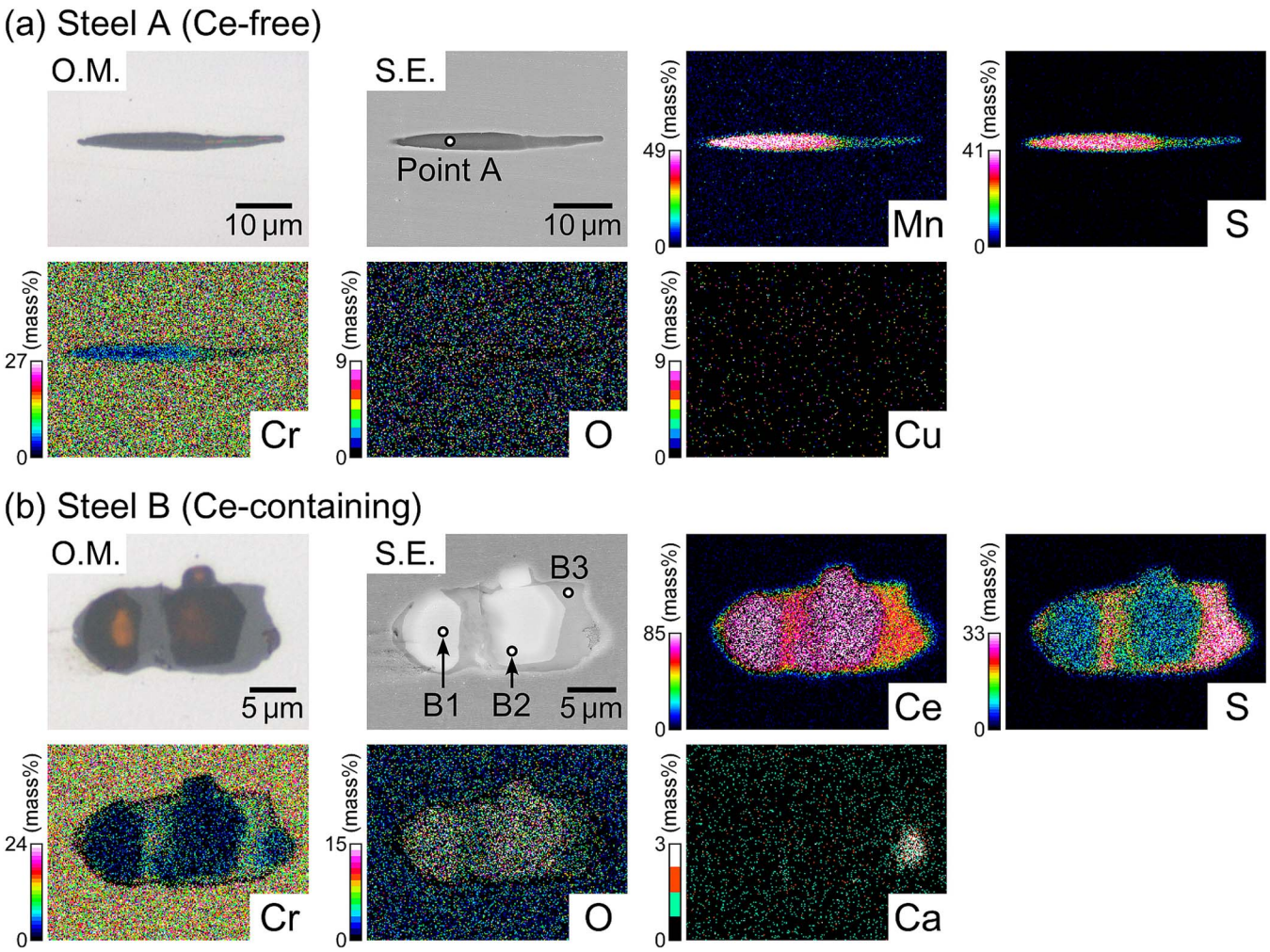


Figure 1. Optical micrographs, SEM images, and EDS maps of inclusions: (a) steel A, (b) steel B.

Cathodic current flowed at first, and then anodic current was measured above approximately −0.03 V. Current spikes which observed in the potential range from ca. 0.3 to 0.41 V are indicative of pit initiation. The large increase in current density at 0.41 V was attributed to stable pit initiation. It is known that pits occur in the dissolution potential range of the MnS inclusions (from ca. 0.3 to 0.5 V).[12–14,16–19] Figure 2b shows the optical micrograph of the MnS before polarization, and Figure 2c shows the SEM image after polarization. Fig. 2c shows the stable pit and meta-stable pits initiated at the boundary between the MnS inclusion and the steel matrix.

FE-SEM/EDS was used to image the dissolution morphology of the inclusion at the pit initiation sites. Figure 3 provides the SEM image and the EDS maps of the MnS inclusion after the polarization shown in Fig. 2. The inclusion surface dissolved slightly, and a trench was formed at the MnS/steel matrix boundary. It has been reported that the formation of trenches at the boundaries leads to the exposure of the bare steel surface and creates an occluded environment.[7–10] The pitting at the MnS inclusions is supposed to be the localized transition from the passive to the active state of the steel matrix inside the trench. The anodic dissolution of the sulfide inclusions in chloride environments is the trigger of pit initiation at the inclusions. While the surface dissolution of the MnS inclusion was observed, a large part of the inclusion remained, as presented in Fig. 3. The EDS maps show that small amounts of Cr, O, and Cu were detected in the inclusion after polarization. The relative atomic percentage at point "C" was Mn:S:Cr:O:Cu = 26:32:11:30:1. Shimahashi et al. demonstrated that the composition of (Mn,Cr)S changes to CrS under anodic polarization, and oxide films form on CrS.[49] The surface oxide films which form on the sulfide inclusions are known to act as a barrier against the dissolution of the inclusion.[25,29,50,51] Lillard et al. proposed that MnS inclusions are passivated by Cu deposition.[21] They analyzed the surface of the MnS inclusion using atomic force microscopy and scanning Kelvin probe force microscopy, and found that Cu deposition on the MnS inclusion causes the MnS ennoblement and passivation during pit propagation.

Table III. Relative concentrations (at.%) of points shown in Fig. 1.

Point	Mn	Ce	S	Cr	O	Ca	Cu	Fe	Ni
A	46	–	45	5	<1	–	<1	4	<1
B1	–	40	15	<1	42	<1	–	3	<1
B2	–	39	17	2	39	<1	–	2	<1
B3	–	29	44	14	5	<1	–	7	1

–: No detection.

Polarization behavior of CeS inclusion.—Micro-scale anodic polarization was carried out to compare the pitting corrosion resistance of the CeS inclusion with that of the MnS inclusion. Figure 4a shows the polarization curve of a small area with the CeS and Ce_2O_2S inclusions in steel B measured in 0.1 M NaCl. Figures 4b and 4c show the optical micrographs of the inclusion before and after polarization, respectively. As shown in Fig. 4b, the CeS inclusion is in the center

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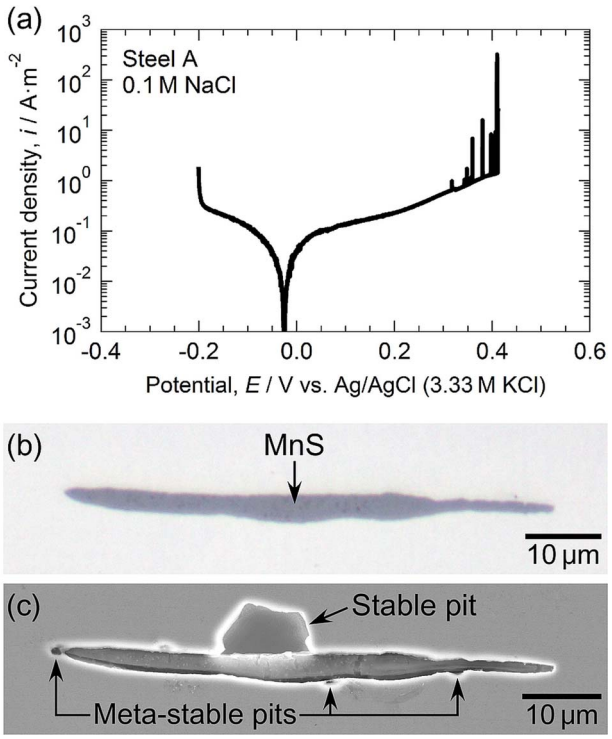


Figure 2. (a) Micro-scale polarization curve of a small area with the MnS inclusion in steel A measured in 0.1 M NaCl at 298 K. (b) Optical micrograph of inclusion before polarization. (c) SEM image of inclusion after polarization.

of the image. As mentioned above (Fig. 1b), CeS was visually distinguishable from Ce_2O_2S in the optical micrograph because of the difference in their surface colors; CeS is light gray, and Ce_2O_2S is orange and dark gray. In Fig. 4a, the polarization curve of a small area with the MnS inclusion in steel A shown in Fig. 2 is given as a reference (the black curve). In steel B, a large current increase was observed at ca. 0.8 V. This was attributed to the stable pit initiation at the inclusion. As shown in Fig. 4c, the surface color of the CeS inclusion changed after polarization, and a pit was generated at the CeS/steel matrix boundary. The pit initiation potential of the CeS inclusion was higher than that of the MnS inclusion, as shown in Fig. 2. However, in both types of inclusions, the pit initiation site was the inclusion/steel matrix boundary. It was suggested that the pitting at the inclusion/steel matrix boundary was attributed to the release of S-species and the exposure of the bare steel surface during inclusion dissolution. In this study, no notable change in the surface of the Ce_2O_2S was observed. This observation was well-correlated with the work of Shimada et al., who demonstrated that Ce-oxysulfides are insoluble in water.[31] In order to ensure reproducibility, polarization in 0.1 M NaCl was repeated three times at different areas with the CeS and Ce_2O_2S inclusions in steel B. Stable and/or meta-stable pits were initiated at approximately 0.8 V in all cases.

FE-SEM/EDS was also used to image the dissolution morphology of the inclusion at the pit initiation site. The SEM image and the EDS maps of the CeS and Ce_2O_2S inclusions in steel B after polarization are shown in Fig. 5a. The polarization curves of these inclusions are shown in Fig. 4. Figure 5b provides the enlarged SEM image of the area surrounded by the dashed lines in Fig. 5a. No dissolution was observed at the surface of the Ce_2O_2S inclusion. Even after anodic polarization, the EDS signals of Ce, S, and O were apparently detected on the Ce_2O_2S inclusions. On the other hand, the surface of the CeS dissolved slightly, which implies that the change in the surface color of the CeS was due to the dissolution and/or corrosion of the inclusion. The trench at the periphery of the CeS inclusion seems to be shallower than that of the MnS inclusion, which is shown in Fig. 3. In terms of the surface dissolution of the inclusion and the pitting at the inclusion/steel

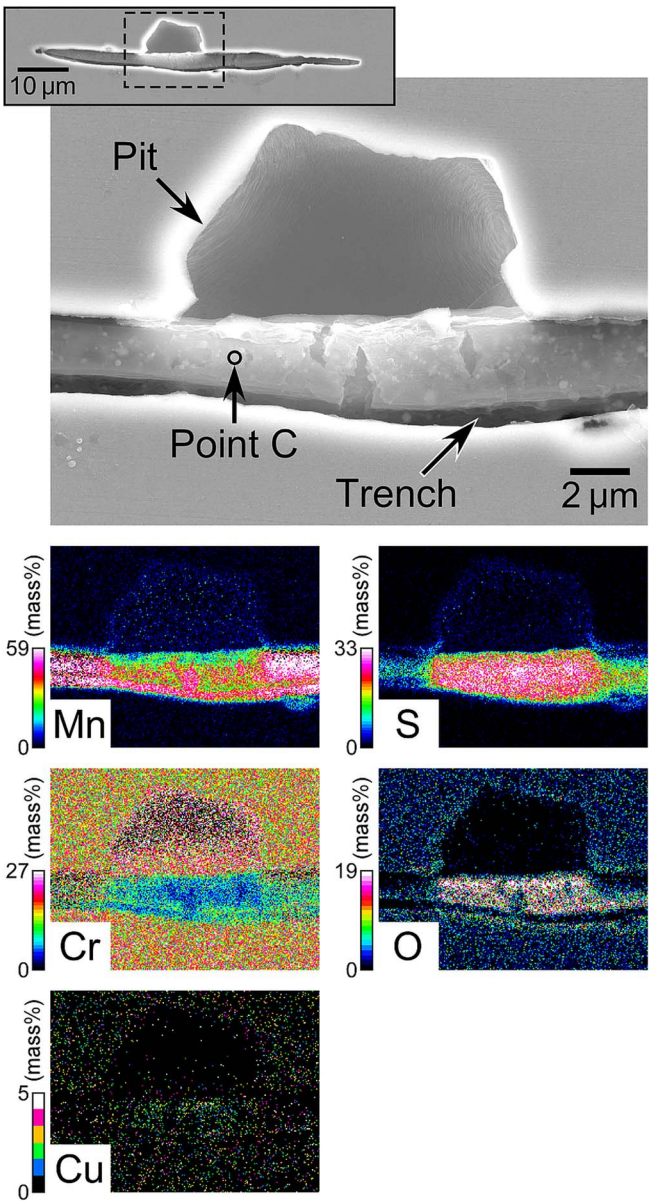


Figure 3. SEM images and EDS maps of inclusion in steel A after the polarization shown in Fig. 2. The inserted SEM image is the whole view of the inclusion, and the area surrounded by dashed lines was enlarged and analyzed.

matrix boundary, the pit initiation behavior at the CeS inclusion was similar to that at the MnS inclusion. However, the pitting corrosion resistance of the CeS inclusion was clearly better than that of the MnS inclusion.

Dissolution behavior of CeS inclusion.—Since the dissolution of the sulfide inclusions is strongly related to pit initiation,[34] it is important to characterize the dissolution behavior of the inclusion. To analyze the anodic dissolution behavior of the CeS inclusion, micro-scale anodic polarization was carried out in 0.1 M Na_2SO_4. Figure 6a shows the polarization curve of a small area with the CeS and Ce_2O_2S inclusions in steel B measured in 0.1 M Na_2SO_4. Figures 6b and 6c show the optical micrographs of the inclusions before and after the polarization. The existence of CeS was confirmed along the periphery of the Ce_2O_2S inclusion. Anodic polarization was started at −0.2 V and was stopped at ca. 1.2 V. In Fig. 6a, the micro-scale polarization curve measured in 0.1 M NaCl, which is shown in Fig. 4, is given as a reference (the black curve). As shown in Fig. 6c, the surface color

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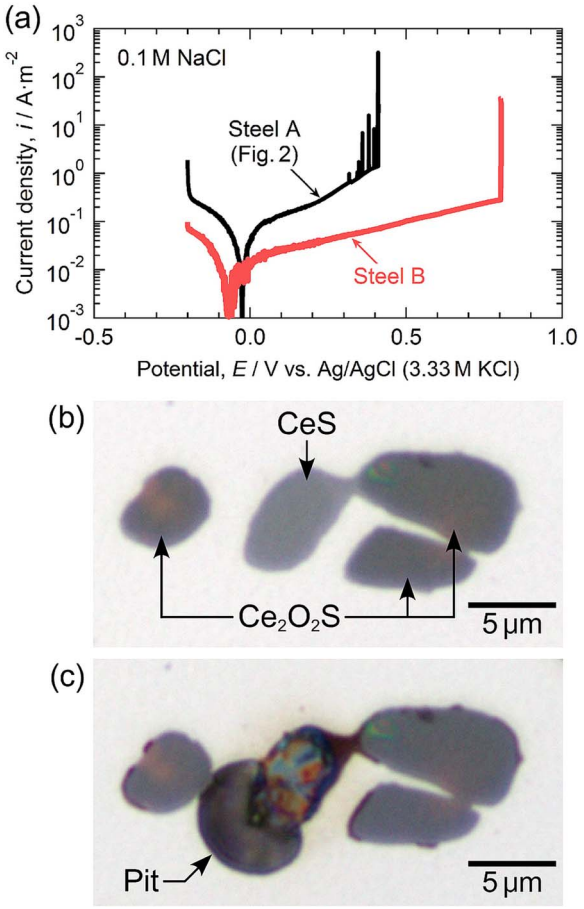


Figure 4. (a) Micro-scale polarization curve of a small area with CeS and Ce_2O_2S inclusions in steel B measured in 0.1 M NaCl at 298 K. Optical micrographs of the inclusions in steel B (b) before and (c) after polarization.

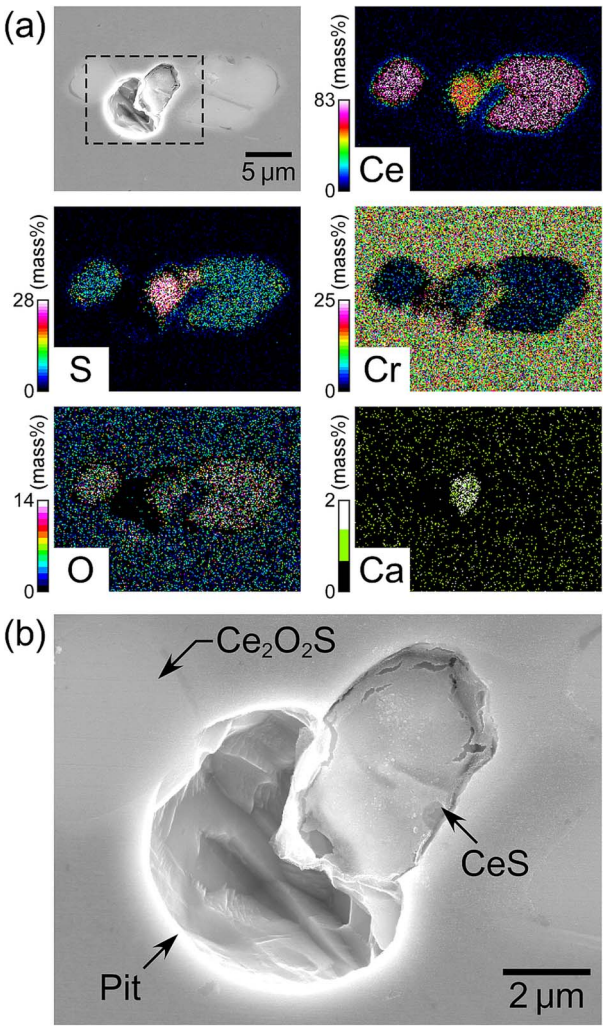


Figure 5. (a) SEM image and EDS maps of an inclusion in steel B after the polarization shown in Fig. 4. (b) Enlarged SEM image of the area surrounded by dashed lines in (a).

of the CeS inclusion changed after polarization. The white corrosion product which appeared at the upper left of the inclusion is thought to be the dissolution product. Figure 7 provides the SEM image and the EDS maps of the inclusion after polarization, shown in Fig. 6. While no dissolution was observed at the surface of the Ce_2O_2S inclusion, the CeS inclusion partially dissolved. The change in the surface color of the CeS inclusion can be attributed to its dissolution and/or corrosion. The EDS maps show that the concentrations of Ce and S decreased as a result of anodic polarization, but the Cr content remained. In addition, the concentration of O increased in the CeS after polarization. It was suggested that the preferential dissolution of Ce and S occurred under anodic polarization. Polarization in 0.1 M Na_2SO_4 was replicated twice at different areas with the CeS and Ce_2O_2S inclusions in steel B. The same dissolution behavior was observed in both cases.

To analyze the dissolution potential of the CeS, in situ observation during anodic polarization was employed. Figure 8 displays the optical micrographs (taken from the video recording) of the electrode surface with the CeS and Ce_2O_2S inclusions during the polarization shown in Fig. 6. As indicated by the arrow in Fig. 8a, the surface of the CeS inclusion was bright-brown when polarization was started at −0.2 V. The inclusion changed in color, becoming dark-blue at −0.1 V (Fig. 8b). After that, no color change was observed from 0.0 to 0.1 V, as shown in Figs. 8c and 8d. The color of the inclusion changed to brown-green at 0.2 V (Fig. 8e) and turned back to dark-blue at 0.3 V (Fig. 8f). Then, the inclusion became brown-red at 0.4 V (Fig. 8g). As shown in Fig. 8h, at 0.5 V, the inclusion turned white, and a white round product appeared at the upper left of the inclusion (marked by the thick arrow). In Figs. 8h and 8i, the white product grew to the left in the potential range from 0.5 to 0.6 V. This white product is thought to be

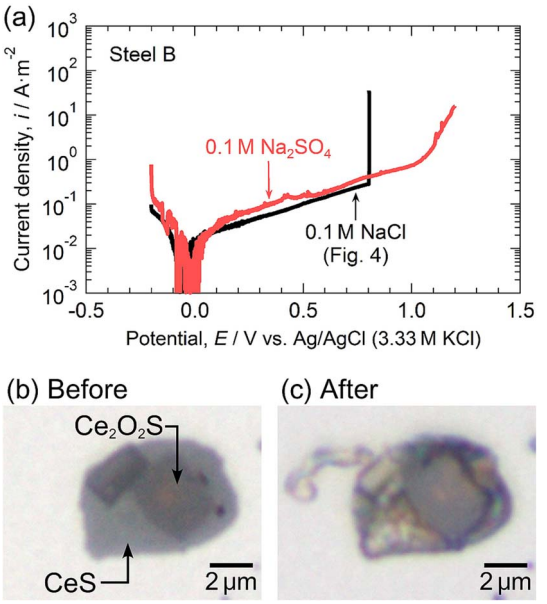


Figure 6. (a) Polarization curve of a small area with CeS and Ce_2O_2S inclusions in steel B measured in 0.1 M Na_2SO_4 at 298 K. Optical micrographs of inclusion (b) before and (c) after polarization in 0.1 M Na_2SO_4.

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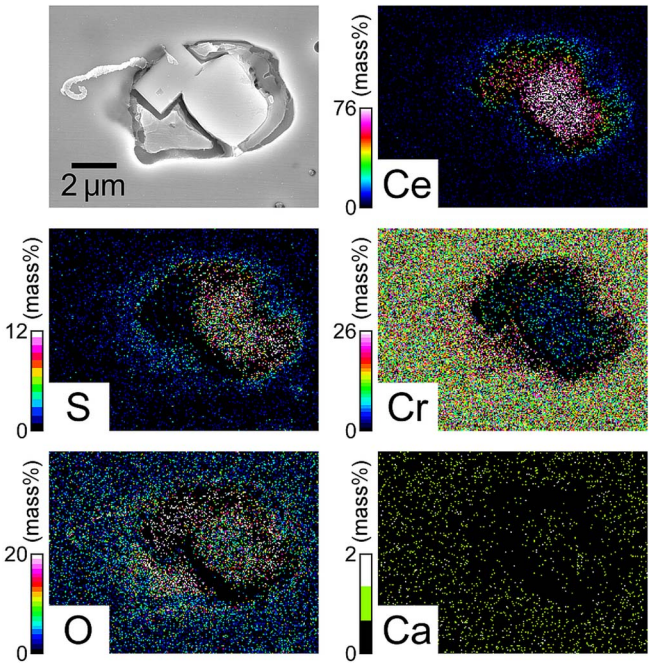


Figure 7. SEM image and EDS maps of the inclusion in steel B after the polarization shown in Fig. 6.

the dissolution product from the CeS inclusion, although its chemical composition was not clear from the SEM/EDS analysis shown in Fig. 7. There was no notable change in the inclusion surface above 0.6 V (Figs. 8i–8o). Therefore, it was suggested that the dissolution of the CeS inclusion occurred at approximately 0.5 V. The color change of the CeS inclusion in the potential range from 0.1 to 0.4 V was thought to be due to the change in the thickness of the surface films.

To determine the dissolution potential of the CeS, additional micro-scale anodic polarization was conducted in 0.1 M Na_2SO_4. Figure 9a shows the polarization curve of a small area with the CeS and Ce_2O_2S inclusions in steel B measured in 0.1 M Na_2SO_4. Figures 9b and 9c show the optical micrograph of the inclusions before polarization and the SEM image after polarization. In this case, the potentiodynamic polarization was stopped at 0.6 V. In this figure, the polarization curves of a small area with the CeS inclusion shown in Figs. 4 and 6 were given as references. The increase in current density at approximately 0.5 V can likely be attributed to the dissolution of the CeS inclusion. The dissolution of the CeS inclusion can clearly be observed in Fig. 9c. Therefore, it was concluded that the dissolution potential of the CeS inclusion is approximately 0.5 V. Polarization in 0.1 M Na_2SO_4 was repeated twice at different areas with the CeS and Ce_2O_2S inclusions in steel B. The same dissolution behavior was observed in both cases.

Interestingly, a comparison of the pitting and the CeS inclusion dissolution potentials (Figs. 4 and 9) indicates that the pit initiation potential at the CeS inclusion was considerably higher than the dissolution potential of the CeS inclusion. In the case of MnS inclusions, it is well-known that stable pitting is initiated in the dissolution potential range of MnS inclusions.[12–14,17,18] MnS dissolution directly results in the formation of a trench around the inclusions, and the meta-stable and stable pitting events proceed after the trench is formed. On the other hand, the pit initiation potential at the CeS inclusion is approximately 0.3 V higher than the dissolution potential. If the NaCl concentration is high enough, pits would occur in the dissolution potential. However, polarization in 0.1 M NaCl at different areas with the CeS and Ce_2O_2S inclusions was replicated three times, and no pitting was initiated in the dissolution potential of the CeS inclusions in all cases. This suggests that the dissolution products from the CeS inclusion inhibit pit initiation, resulting in the increase of the pit initiation potential at the CeS inclusion.

Potential-pH diagrams for the CeS-H_2O system.—To determine the dissolution products from the CeS inclusion, the potential-pH diagrams were calculated. Figure 10 shows the potential-pH diagrams for the CeS-H_2O system at 298 K. The standard chemical potentials of $Ce(OH)_2^{2+}$, CeO_2, $CeOH^{2+}$, and $Ce(OH)_3$ were obtained from the literature.[52,53] Those of other species used in the calculations were obtained from the HSC thermo-chemical database.[54] The concentration of the soluble species in the calculation was 1.0×10^{-3} mol kg^{-1} (H_2O). The experimental conditions of the anodic polarization in 0.1 M NaCl (Fig. 4a) are indicated by the slashed rectangles located in the stable regions of $S_2O_3^{2-}$, Ce^{3+}, and CeO_2. The stable region of CeS is located in the lower potential region. These diagrams indicate that Ce^{3+} and CeO_2 were likely formed during anodic polarization in 0.1 M NaCl. As shown in Fig. 10b, Ce^{3+} ions were released from the CeS

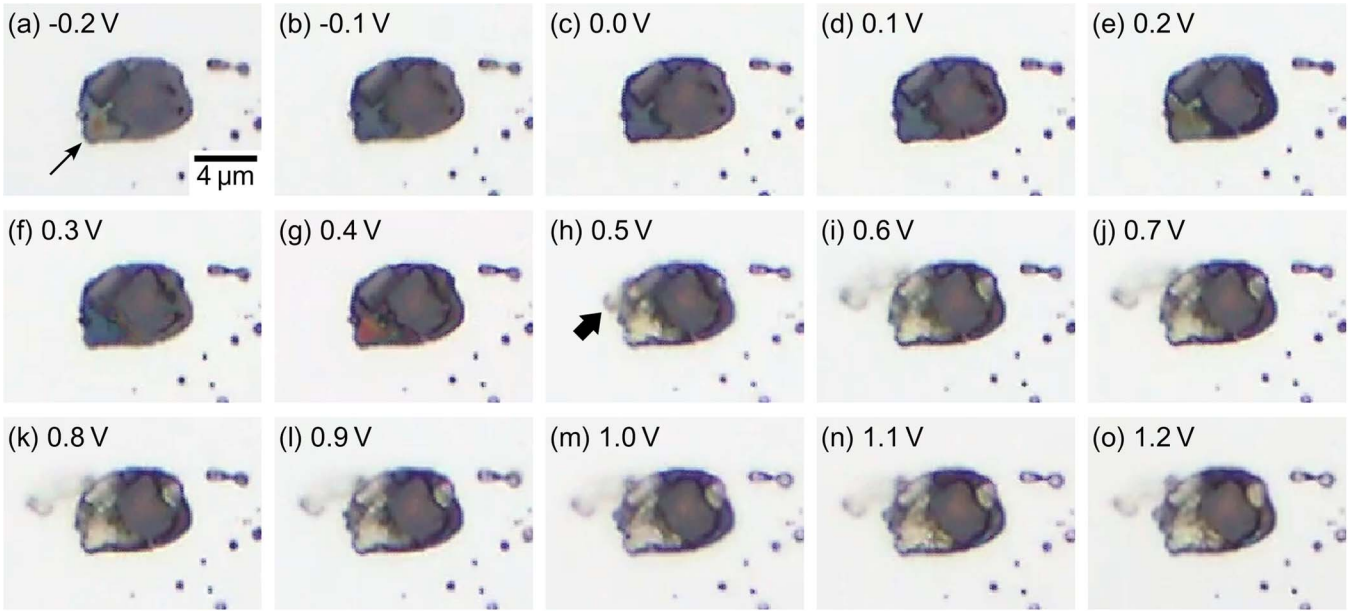


Figure 8. Optical micrographs of electrode surface with CeS and Ce_2O_2S inclusions in steel B during the polarization shown in Fig. 6.

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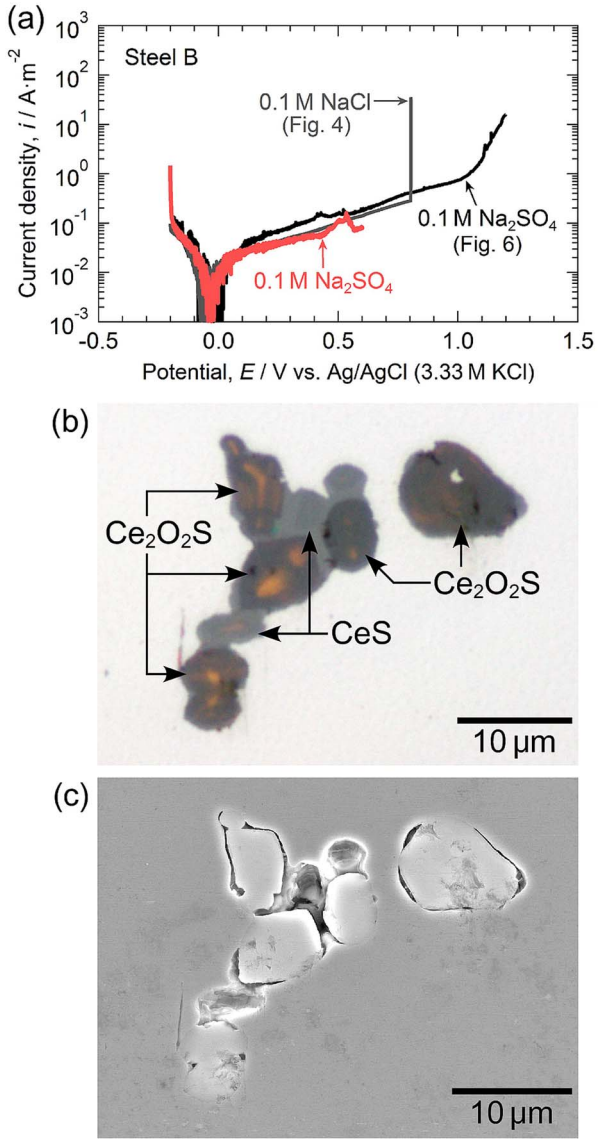


Figure 9. (a) Polarization curve of a small area with CeS and Ce_2O_2S inclusions in steel B measured in 0.1 M Na_2SO_4 at 298 K. The polarization was stopped at 0.6 V. (b) Optical micrograph of an inclusion before polarization. (c) SEM image of an inclusion after polarization in 0.1 M Na_2SO_4.

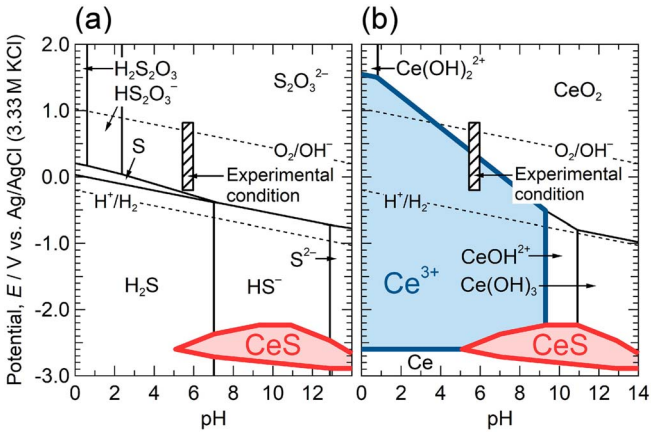


Figure 10. Potential-pH diagrams for CeS-H_2O system at 298 K. The concentration of soluble species is 1.0×10^{-3} mol kg^{-1} (H_2O): (a) CeS and S-species, (b) CeS and Ce-species.

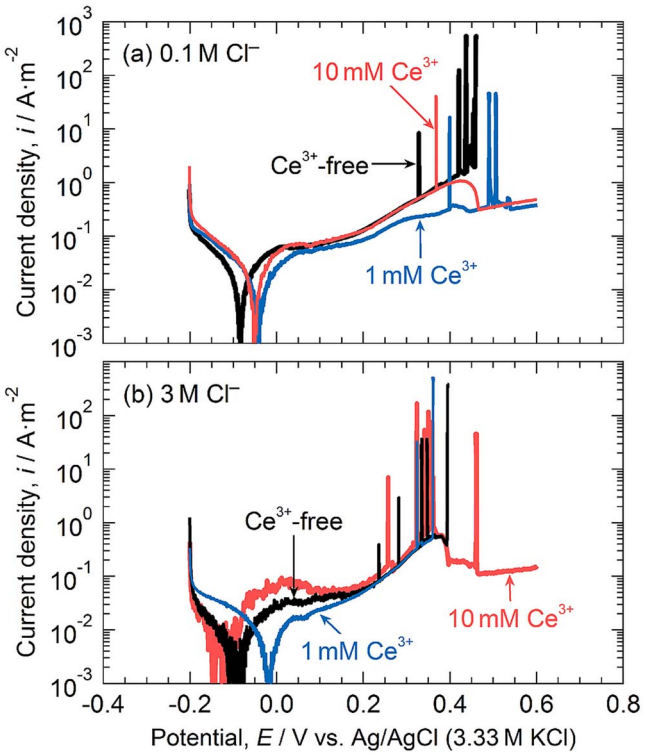


Figure 11. Micro-scale polarization curves of a small area with the MnS inclusion in steel A measured in NaCl-$CeCl_3$ solutions at 298 K: (a) 0.1 M Cl^-, (b) 3 M Cl^-.

inclusion at lower potentials during anodic polarization. At higher potentials, CeO_2 was apparently deposited on the CeS inclusion, suppressing the dissolution of the inclusion. In this study, the inhibition effect of Ce^{3+} ions on pit initiation was intended to be examined.

Effect of Ce^{3+} on pitting corrosion resistance of MnS inclusion.—The data presented above suggest that the dissolving Ce^{3+} ions play an important role in the inhibition of pitting corrosion at CeS inclusions. Assuming that Ce^{3+} ions inhibit pitting corrosion at sulfide inclusions, pitting at MnS inclusions would also be suppressed in the Ce^{3+}-containing solutions. To elucidate the effect of Ce^{3+} ions on the pitting corrosion resistance of MnS inclusions, micro-scale polarization curves of a small area with the MnS inclusion in steel A were measured in the NaCl-$CeCl_3$ solutions. The area studied contained one large MnS inclusion positioned at the center. The electrolytes were 0.1 M NaCl, 3 M NaCl, and the NaCl-$CeCl_3$ solutions. Figure 11 shows the micro-scale anodic polarization curves. Anodic polarization was started at −0.2 V. The gradual increase in current density in the potential range from ca. 0.2 V to 0.5 V was attributed to the dissolution of the MnS inclusions. The current spikes indicate pit initiation. In the case of the 0.1 M Cl^- solutions, a stable pit was generated at 0.46 V in the Ce^{3+}-free solution. On the other hand, the pits were re-passivated and the current density decreased at approximately 0.45 V in the Ce^{3+}-containing solutions. In the case of the 3 M Cl^- solutions, stable pits were generated at approximately 0.4 V in the Ce^{3+}-free and 1 mM Ce^{3+} solutions. However, no stable pit was initiated and the current density decreased at approximately 0.4 V in the 10 mM Ce^{3+} solution. Clearly, Ce^{3+} ions inhibited the stable pit initiation at the MnS inclusions. It can therefore be concluded that the Ce^{3+} ions released from CeS inclusions improve the pitting corrosion resistance of these inclusions.

Role of Ce^{3+} for improving pitting corrosion resistance of sulfide inclusion.—As mentioned above (Fig. 11), pitting at the MnS inclusions was suppressed in the Ce^{3+}-containing solutions. The Ce^{3+} ions

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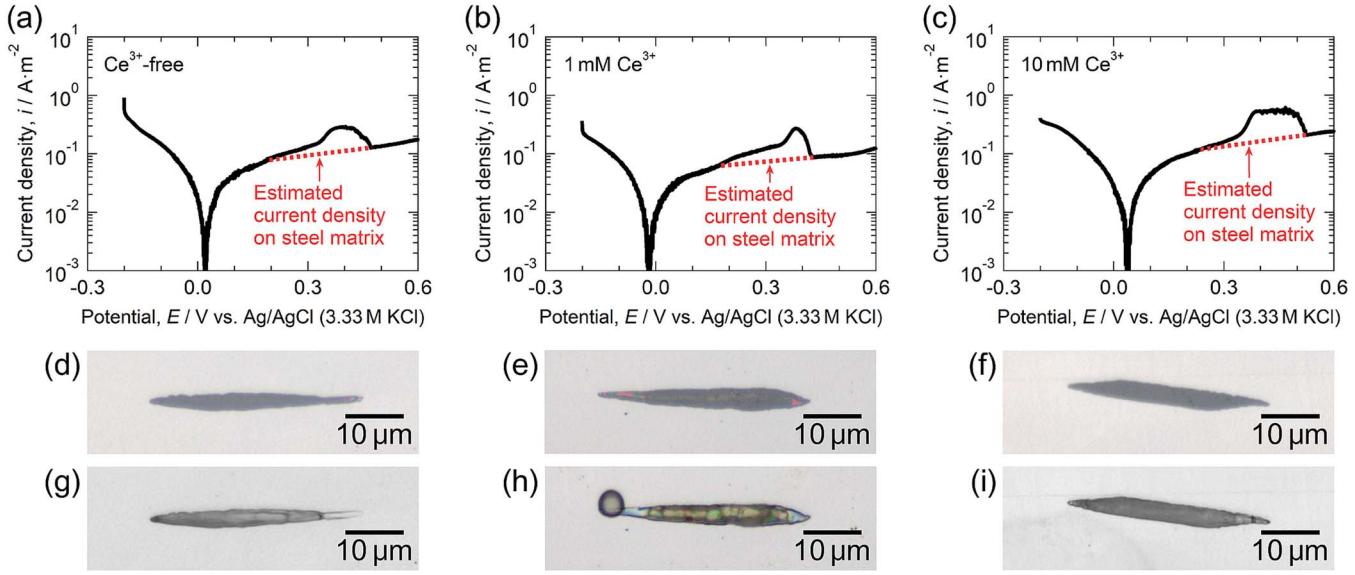


Figure 12. Micro-scale polarization curves of a small area with the MnS inclusion in steel A measured in the Na_2SO_4-$Ce_2(SO_4)_3$ solution at 298 K: (a) Ce^{3+}-free, (b) 1 mM Ce^{3+}, (c) 10 mM Ce^{3+}. Optical micrographs of MnS inclusions in steel A (d–f) before and (g–i) after polarization: (d, g) Ce^{3+}-free, (e, h) 1 mM Ce^{3+}, (f, i) 10 mM Ce^{3+}.

have very likely inhibited one of the steps in the pit initiation mechanism at MnS inclusions. Chiba et al.[19] proposed that the pit initiation mechanism at MnS inclusions in stainless steels proceeds as follows: 1) the dissolution of MnS inclusions in chloride solutions leads to the release of thiosulfate ions and weak acidification; 2) elemental sulfur is deposited on and around the inclusions through the disproportionation reaction and the reduction reaction of the thiosulfate ions in acidic solution; 3) the synergistic effect of the elemental sulfur and chloride ions causes trench formation at the inclusion/steel matrix boundary; 4) pits are initiated when the solution inside the trench reaches the critical conditions of the local active dissolution of the steel matrix. On the basis of this pit initiation mechanism, the role of Ce^{3+} ions for improving the pitting corrosion resistance of MnS inclusions was evaluated.

To analyze the effect of Ce^{3+} ions on the extent of the dissolution of MnS inclusions, the micro-scale anodic polarization of a small area with a MnS inclusion was performed. Figures 12a–12c show the polarization curves measured in the Na_2SO_4-$Ce_2(SO_4)_3$ solutions. Figures 12d–12f and 12g–12i show the optical micrographs of the MnS inclusions before and after polarization, respectively. Polarization was started at −0.2 V and was stopped at 0.6 V. The broad current peak at approximately 0.3 V in each of the polarization curves indicates the dissolution of the MnS inclusions.[12,13] The current densities on the steel matrix were estimated by the dashed lines in Figs. 12a–12c. The charge passed owing to MnS dissolution was calculated after subtracting the current on the steel matrix from the total current, and then normalizing by the surface area of the MnS inclusion. The surface areas of the MnS inclusions were measured by digital image processing of confocal laser scanning microscopy data before polarization. Table IV shows the calculated electric charges, the surface area of the MnS inclusions, and the electrode area.

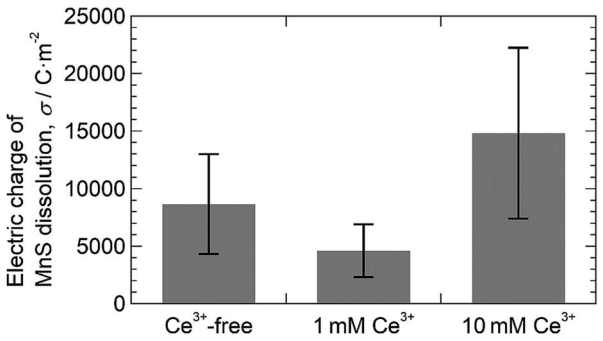


Figure 13. Electric charges of MnS dissolution per unit MnS inclusion area measured in the micro-scale polarization shown in Fig. 12.

Figure 13 shows the normalized charge passed from MnS dissolution calculated from the polarization curves measured in the Ce^{3+}-free, 1 mM Ce^{3+}, and 10 mM Ce^{3+} solutions. Polarization in the Na_2SO_4-$Ce_2(SO_4)_3$ solutions was not replicated. We consider that the charge passed by inclusion dissolution has a wide range of error due to the difference in the shape, the size, and the chemical composition of the MnS inclusions used in this experiment. The error bars shown in Fig. 13 indicated that the charge passed by inclusion dissolution has an error within a factor of two. It was suggested that Ce^{3+} addition did not suppress the dissolution of the MnS inclusions. As seen in Figs. 5, 7, and 9, the dissolution of the CeS inclusions was observed while the Ce^{3+} ions was thought to exist near the inclusions.

To confirm whether the Ce^{3+} ions have a pH buffering effect, the titration curves of the NaCl-$CeCl_3$ solutions were measured. Figure 14

Table IV. Dissolution amount of MnS inclusion calculated from the polarization curve in Fig. 12.

Solution	Ce^{3+}-free	1 mM Ce^{3+}	10 mM Ce^{3+}
Electrode area (μm^2)	1.21×10^4	1.06×10^4	8.53×10^3
Electric charge of MnS dissolution (μC)	0.71	0.47	1.37
Surface area of MnS inclusion (μm^2)	82.0	102	92.5
Electric charge of MnS dissolution per unit area (C m^{-2})	8.66×10^3	4.61×10^3	1.48×10^4

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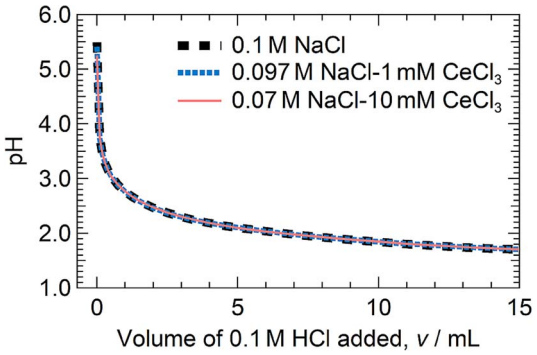


Figure 14. Titration curves of the NaCl and NaCl-$CeCl_3$ solutions.

shows the change in the solution pH values of the Ce^{3+}-free, 1 mM Ce^{3+}, and 10 mM Ce^{3+} solutions with the addition of 0.1 M HCl. The pH values of the solutions decreased with HCl addition. As seen in Fig. 14, no difference was found in these titration curves. This confirms that Ce^{3+} ions do not act as a pH buffer.

It has been reported that the trench morphology at the MnS/steel matrix boundary is strongly related to the pitting corrosion resistance.[12,55,56] To analyze the effect of Ce^{3+} ions on trench formation at the MnS/steel boundary, the micro-scale anodic polarization curves of a small area with a MnS inclusion were measured in 3 M NaCl and 2.97 M NaCl-10 mM $CeCl_3$. Figure 15a shows the polarization curves measured in 3 M NaCl and 2.97 M NaCl-10 mM $CeCl_3$. Polarization was started at −0.2 V and was stopped at 0.37 V. A stable pit occurred in the Ce^{3+}-free solution, and no stable pit was initiated in the Ce^{3+}-containing solution. After polarization, the electrode surfaces and the cross-sections of the MnS inclusions were observed. Figures 15b–15e show the SEM images of the MnS inclusions after polarization. As shown in Figs. 15b and 15d, in the Ce^{3+}-free solution, the boundary between the MnS inclusion and the steel matrix dissolved selectively, and the deep trenches were formed. In contrast, in the Ce^{3+}-containing solution (Figs. 15c and 15e), either a shallow trench or no trench was formed. It was found that Ce^{3+} ions inhibited the formation of a deep trench at the MnS/steel matrix boundary. Polarization in the Ce^{3+}-containing solution was repeated twice at different areas with the MnS inclusions in steel A. In both cases, no stable pit was initiated until 0.37 V and the inhibition effect of Ce^{3+} ions on the formation of a deep trench was observed.

Figure 16 (bottom) shows the schematic of the pit initiation mechanism at CeS inclusions. The pit initiation mechanism at MnS inclusions proposed by Chiba et al.[19] is also shown in Fig. 16 (top) for comparison. In the case of MnS inclusions, the dissolution of the inclusion causes the formation of a deep trench at the inclusion/steel matrix boundary, and a stable pit tends to occur at the bottom of the deep trench.[18] On the other hand, in the case of CeS inclusions, Ce^{3+} ions are released from the inclusion during anodic polarization. The released Ce^{3+} ions inhibit the formation of a deep trench at the inclusion/steel matrix boundary. The composition of the solution inside the shallow trench around the CeS inclusion is assumed to be weak compared to that around the MnS inclusion. The superior pitting corrosion resistance of CeS inclusions compared to that of MnS inclusions can

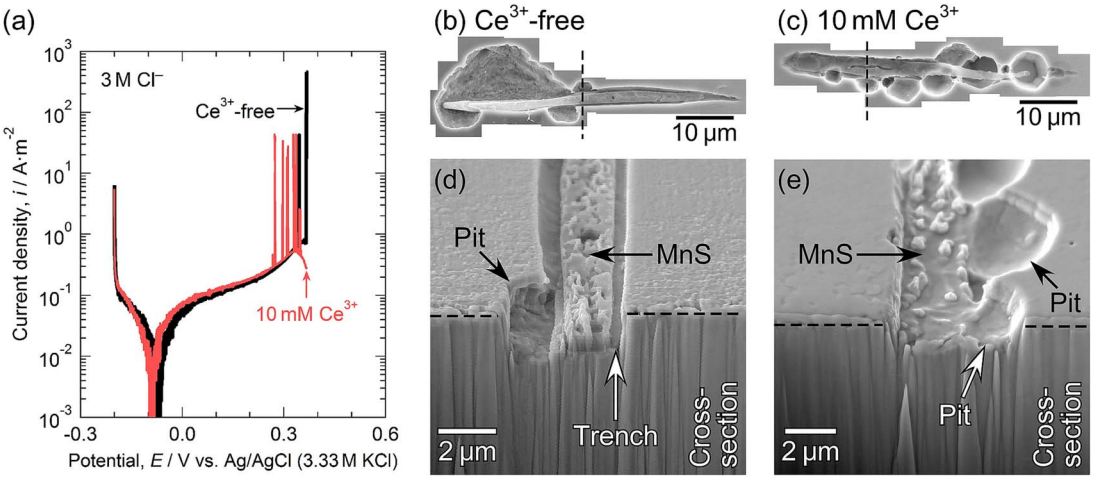


Figure 15. (a) Micro-scale polarization curves of a small area with the MnS inclusion in steel A measured in 3 M NaCl and 2.97 M NaCl-10 mM $CeCl_3$ at 298 K. SEM images of the MnS inclusion after polarization measured in (b) 3 M NaCl and (c) 2.97 M NaCl-10 mM $CeCl_3$; the accelerating voltage was 20 kV. (d, e) SEM images of FIB cross-sections at the sites indicated by the dashed lines shown in (b, c); the accelerating voltage was 10 kV.

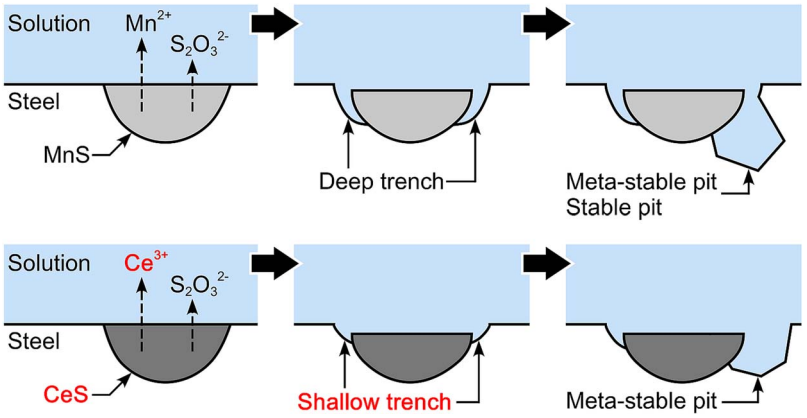


Figure 16. Schematic of pit initiation mechanisms at (a) MnS and (b) CeS inclusions in NaCl solutions.

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be thus explained. However, the mechanism of the inhibition effect of Ce^{3+} on trench formation remains unclear. One possibility is that the Ce^{3+} ions form a complex with Cl^- ions and prevent the attack of Cl^- ions against the steel matrix. In the field of automotive emissions control, it has been reported that Ce compounds behave as sorbents for the removal of S from fuel and flue gases.[57] This suggests that it is possible that, in the presence of Ce^{3+} ions, the S released from the inclusions is not all deposited. Further studies are required to determine the inhibition mechanism of Ce^{3+} on dissolution at the inclusion/steel matrix boundary.

Conclusions

1. A stable pit was generated at the CeS inclusion at 0.8 V in naturally aerated 0.1 M NaCl (298 K). The pitting potential at the CeS inclusion was 0.4 V higher than that of the MnS inclusion. However, the pit initiation site, which was the inclusion/steel matrix boundary, was found to be common to both these inclusions.
2. In the case of the MnS inclusion, a stable pit was initiated in the potential range of the inclusion dissolution. The dissolution of the CeS inclusion proceeded at approximately 0.5 V, but pit initiation was suppressed until 0.8 V. The difference in the pitting and dissolution potential can likely be attributed to the inhibitive effect of Ce-species on pitting. On the basis of thermodynamic calculations, it is suggested that the Ce^{3+} ions were released from CeS inclusions.
3. The Ce^{3+} ions improved the pitting corrosion resistance at the MnS inclusion in 0.1 M and 3 M NaCl solutions. In the microscale polarization of a small area with a MnS inclusion, a stable pit was initiated in Ce^{3+}-free solutions, whereas no stable pit was generated in Ce^{3+}-containing solutions even when metastable pitting events were observed. In the Ce^{3+}-free solution, the boundary between the MnS inclusion and the steel matrix dissolved selectively, and a deep trench was formed. On the other hand, only a shallow trench was formed in the Ce^{3+}-containing solution. This suggests that the Ce^{3+} ions inhibit dissolution at the MnS/steel boundary, resulting in the improved pitting corrosion resistance of sulfide inclusions in stainless steels.

Acknowledgments

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